

# Solar Industry Expected to Grow Fast

In spite of a few challenges, the Indian solar industry is expected to grow substantially from 2600MW to 20,000MW by 2022 with the help of government and solar manufacturers' initiatives. We might soon see Indian cities turning into 'solar cities' in the near future

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**T**he solar power industry is set to turn around quickly and establish its place amongst the topmost industries in India. With a total worldwide installed capacity of 136,697MW, this industry has recorded a growth of almost 35 per cent. And India has started contributing to it. Recently, the Renewable Energy Policy Network for the 21st Century (Ren21) ranked India amongst the top five nations worldwide for Concentrated Solar Power (CSP) technology capacity.

It is assumed that India would add a 50MW plant on a 1.27 million square metre floating platform by the end of this year. Currently, India's solar capacity is 2600MW only, but it is targeted to cross 20GW by 2022. It is said that India can generate 1.5 times the current electricity demand with just 1000GW of solar power. In order to build it, India needs just 0.5% of its land, or half the desert district of Barmer in Rajasthan.

The Commerce Ministry has recommended anti-dumping duty of US\$0.11 to US\$0.81 per watt to secure the market around of domestic manufacturers. A latest report by Powergrid suggests, "Market analysis has broad consensus that utility-scale solar photo-voltaic (PV) generation will achieve grid parity around 2016-17, based on the pace of reduction in cost of PV technology and rise in conventional energy prices."

## The peculiar situation in India

India is in a similar situation as Japan when it comes to energy. Both these nations have very little land in places



Fig. 1: Solar power

where solar power plants need to be installed. This sounds very contradictory, specially when we see areas like Leh, Rajasthan, Gujarat and Madhya Pradesh. We have plenty of land available there.

Here is the twist to this statement: There is deficiency of power in places like Bihar, Uttar Pradesh, Nashik, Tamil Nadu and Kerala, which have limited land for the installation of huge solar power plants. These places also lack the transmission infrastructure. To this, Sandip Ghosh, EVP-solar products, Jakson Group says, "You cannot transmit power from Rajasthan to Tamil Nadu or to Bihar. The transmission and distribution losses will be close to 10%, and you will also require a huge investment for transmitting this type of power to the end consumer. So what is thought is instead of trying to get into captive power, you get into distributed power generation." He explains, "In places

like Bihar, where we have less cultivated land but abundant water bodies, floating plants are proposed." We have about 10MW of solar plants over canals now, and this number includes 1MW solar power plant that was put up in Gujarat about a year ago.

Sandip adds, "This has several benefits. It enables utilisation of water bodies where there is scarcity of land, reduces the evaporation of the water, and size of the water body below the plant remains cool, and its size does not get affected."

S.P. Gon Choudhury, chairman of Renewable Energy College says, "There are large stretches of water bodies in Kerala which NHPCL (a local energy company) wants to harness for solar power. This floating solar power technology was developed by the Renewable Energy College and has been implemented in the city." He adds, "The first plant (a pilot project) is scheduled to be commissioned in